Assignment 5

OPTI 544 Quantum Optics

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Exercise 1

a) The eigenvalues $E_\pm$ are:

$$
\begin{aligned}
|IE_\pm - \tilde{H}_{\text{eff}}| &= 0 \\
\left| \begin{bmatrix} E_\pm & 0 \\ 0 & E_\pm \end{bmatrix} - \hbar \begin{bmatrix} -\Delta_L/2 & \Omega/2 \\ \Omega^*/2 & \Delta_L/2 \end{bmatrix} \right| &= \\
\begin{vmatrix} \frac{2E_\pm + \hbar\Delta_L}{2} & -\frac{\hbar\Omega}{2} \\ -\frac{\hbar\Omega^*}{2} & \frac{2E_\pm - \hbar\Delta_L}{2} \end{vmatrix} &= \\
(2E_\pm + \hbar\Delta_L)(2E_\pm - \hbar\Delta_L) - \hbar^2|\Omega|^2 &= 0 \\
4E_\pm^2 - \hbar^2\Delta_L^2 - \hbar^2|\Omega|^2 &= \\
E_\pm &= \pm\frac{\hbar}{2}\sqrt{\Delta_L^2 + |\Omega|^2} = \pm\frac{\hbar\Omega_R}{2}.
\end{aligned}
$$

The eigenvectors $v_\pm$ are

$$
\begin{aligned}
\tilde{H}_{\text{eff}} v_\pm &= E_\pm v_\pm \\
\hbar \begin{bmatrix} -\Delta_L/2 & \Omega/2 \\ \Omega^*/2 & \Delta_L/2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \pm\frac{\hbar\Omega_R}{2} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\
\begin{bmatrix} -\frac{\Delta_L}{2}v_1 + \frac{\Omega}{2}v_2 \\ \frac{\Omega^*}{2}v_1 + \frac{\Delta_L}{2}v_2 \end{bmatrix} &= \begin{bmatrix} \pm\frac{\Omega_R}{2}v_1 \\ \pm\frac{\Omega_R}{2}v_2 \end{bmatrix} \longrightarrow v_2 = \frac{\Delta_L \pm \Omega_R}{\Omega} v_1 \Longrightarrow v_\pm = \begin{bmatrix} 1 \\ \frac{\Delta_L \pm \Omega_R}{\Omega} \end{bmatrix},
\end{aligned}
$$

or, by normalizing,

$$
v_\pm = \frac{1}{\sqrt{2\Omega_R(\Omega_R \pm \Delta_L)}} \begin{bmatrix} \Omega \\ \Delta_L \pm \Omega_R \end{bmatrix} = \frac{1}{N_\pm} \begin{bmatrix} \Omega \\ \Delta_L \pm \Omega_R \end{bmatrix}.
$$

The eigenvector matrix is:

$$
S = \begin{bmatrix} | & | \\ v_+ & v_- \\ | & | \end{bmatrix} = \begin{bmatrix} \Omega/N_+ & \Omega/N_- \\ (\Delta_L + \Omega_R)/N_+ & (\Delta_L - \Omega_R)/N_- \end{bmatrix} = V^{-1}.
$$

Then,

$$
\begin{aligned}
S^{-1}\tilde{H}_{\text{eff}}S &= \hbar \begin{bmatrix} \frac{\Omega}{N_+} & \frac{\Omega}{N_-} \\ \frac{\Delta_L+\Omega_R}{N_+} & \frac{\Delta_L-\Omega_R}{N_-} \end{bmatrix}^{-1} \begin{bmatrix} -\Delta_L/2 & \Omega/2 \\ \Omega^*/2 & \Delta_L/2 \end{bmatrix} \begin{bmatrix} \frac{\Omega}{N_+} & \frac{\Omega}{N_-} \\ \frac{\Delta_L+\Omega_R}{N_+} & \frac{\Delta_L-\Omega_R}{N_-} \end{bmatrix} \\
&= \frac{-\hbar N_+ N_-}{2\Omega\Omega_R} \begin{bmatrix} \frac{\Delta_L-\Omega_R}{N_-} & -\frac{\Omega}{N_-} \\ -\frac{\Delta_L+\Omega_R}{N_+} & \frac{\Omega}{N_+} \end{bmatrix} \begin{bmatrix} -\Delta_L/2 & \Omega/2 \\ \Omega^*/2 & \Delta_L/2 \end{bmatrix} \begin{bmatrix} \frac{\Omega}{N_+} & \frac{\Omega}{N_-} \\ \frac{\Delta_L+\Omega_R}{N_+} & \frac{\Delta_L-\Omega_R}{N_-} \end{bmatrix} \\
&= \frac{-\hbar N_+ N_-}{4\Omega\Omega_R} \begin{bmatrix} \frac{\Delta_L-\Omega_R}{N_-} & -\frac{\Omega}{N_-} \\ -\frac{\Delta_L+\Omega_R}{N_+} & \frac{\Omega}{N_+} \end{bmatrix} \begin{bmatrix} \frac{\Omega_R\Omega}{N_+} & -\frac{\Omega_R\Omega}{N_-} \\ \frac{|\Omega|^2+\Delta_L^2+\Delta_L\Omega_R}{N_+} & \frac{|\Omega|^2+\Delta_L^2-\Delta_L\Omega_R}{N_-} \end{bmatrix} \\
&= -\frac{\hbar N_+ N_-}{4\Omega\Omega_R} \begin{bmatrix} \frac{-2\Omega\Omega_R^2}{N_+N_-} & 0 \\ 0 & \frac{2\Omega\Omega_R^2}{N_+N_-} \end{bmatrix} \\
S^{-1}\tilde{H}_{\text{eff}}S &= \begin{bmatrix} \frac{\hbar\Omega_R}{2} & 0 \\ 0 & -\frac{\hbar\Omega_R}{2} \end{bmatrix} = \begin{bmatrix} E_+ & 0 \\ 0 & E_- \end{bmatrix},
\end{aligned}
$$

confirming the correct eigenmodes. Also we need $S^{-}1$ for next parts:

$$
V = S^{-1} = \frac{-N_+N_-}{2\Omega\Omega_R} \begin{bmatrix} \frac{\Delta_L-\Omega_R}{N_-} & -\frac{\Omega}{N_-} \\ -\frac{\Delta_L+\Omega_R}{N_+} & \frac{\Omega}{N_+} \end{bmatrix}.
$$

b) The graph if $E_\pm$ versus Δ_L is shown below.

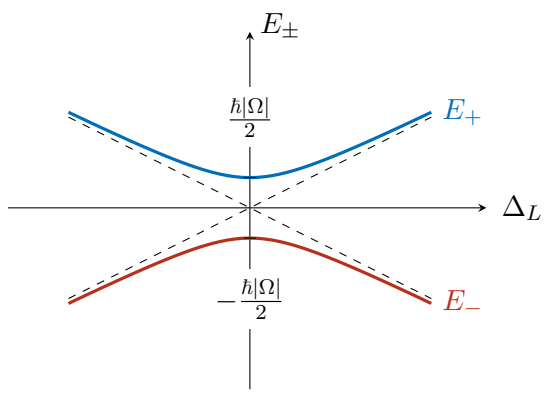


Figure 1: Phase space of the squeezed displaced state for $r = 0.6$.

c) When diagonalizing I have two basis

$$
[\tilde{H}_{\text{eff}}]_{\{|e\rangle,|g\rangle\}} \xrightarrow{\text{Diagonalization}} [\tilde{H}_{\text{eff}}]_{\{|+\rangle,|-\rangle\}},
$$

they are related through

$$
\boldsymbol{C}_{\{|+\rangle,|-\rangle\}} = V\boldsymbol{c}_{\{|e\rangle,|g\rangle\}}, \quad \boldsymbol{C} = \begin{bmatrix} C_+ \\ C_- \end{bmatrix}, \ \boldsymbol{c} = \begin{bmatrix} \tilde{c}_e \\ \tilde{c}_g \end{bmatrix}.
$$

The dressed states $|\pm\rangle$ in the bare basis are related through the eigenvector matrix V^{-1}:

$$
[|+\rangle \ |-\rangle] = [|e\rangle \ |g\rangle]V^{-1} \longrightarrow |\pm\rangle = \frac{1}{N_\pm}[\Omega\,|e\rangle + (\Delta_L \pm \Omega_R)\,|g\rangle], \quad N_\pm = \sqrt{2\Omega_R(\Omega_R \pm \Delta_L)}.
$$

We can get a standard form of the dressed states starting from the coefficient c_e and c_g:

$$c_e^2 = \frac{\Omega^2}{2\Omega_R(\Omega_R+\Delta_L)} = \frac{(\Omega_R+\Delta_L)(\Omega_R-\Delta_L)}{2\Omega_R(\Omega_R+\Delta_L)} = \frac{\Omega_R-\Delta_L}{2\Omega_R} = \frac{1-\Delta_L/\Omega_R}{2},$$
$$c_g^2 = \frac{(\Omega_R+\Delta_L)^2}{2\Omega_R(\Omega_R+\Delta_L)} = \frac{\Omega_R+\Delta_L}{2\Omega_R} = \frac{1+\Delta_L/\Omega_R}{2}.$$

If we equate the last result of c_e^2 to $\cos^2(\theta)$, we get:

$$\cos^2\theta = \frac{1+\cos 2\theta}{2} = \frac{1-\Delta_L/\Omega_R}{2} \Longrightarrow \cos 2\theta = -\frac{\Delta_L}{\Omega_R}.$$

Then,

$$\cos^2 2\theta + \sin^2 2\theta = 1 \longrightarrow \sin 2\theta = \sqrt{1-\Delta_L^2/\Omega_R^2} = \sqrt{\frac{\Omega_R^2-\Delta_L^2}{\Omega_R^2}} = +\frac{\Omega}{\Omega_R}.$$

Using both result we get

$$\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta} = \frac{\Omega/\Omega_R}{-\Delta_L/\Omega_R} = -\frac{\Omega}{\Delta_L}.$$

Therefore,

$$|\pm\rangle = \cos\theta\,|e\rangle + \sin\theta\,|g\rangle\,, \quad \tan 2\theta = -\frac{\Omega}{\Delta_L},$$

and doing the same process we get:

$$|-\rangle = -\sin\theta\,|e\rangle + \cos\theta\,|g\rangle\,.$$

The detuning Δ_L change the amplitude of the bare states. In particular,

$$|\pm\rangle = \begin{cases} |{}^{g}_{e}\rangle\,, & \Delta_L \ll 0 \\ \frac{1}{\sqrt{2}}[|e\rangle \pm |g\rangle], & \Delta_L = 0 \\ |{}^{e}_{g}\rangle\,, & \Delta_L \gg 0 \end{cases}.$$

d) For $\Delta_L = 0$, the dressed states take the form:

$$|\pm\rangle = \frac{1}{\sqrt{2}}[|e\rangle \pm |g\rangle].$$

If the state of the atom is $|\Psi(t)\rangle$, the given initial state is

$$|\Psi(0)\rangle = \frac{1}{\sqrt{2}}[|e\rangle + |g\rangle] = |+\rangle\,.$$

Its evolution in general, will be stationary because $|+\rangle$ is one eigenstate of the Hamiltonian:

$$|\Psi(t)\rangle = e^{-i\Omega t/2}\,|+\rangle\,.$$

At $t = \pi/\Omega$, we have:

$$|\Psi(\pi/\Omega)\rangle = e^{-i\pi/2}\,|+\rangle = -i\,|+\rangle = \frac{-i}{\sqrt{2}}[|e\rangle + |g\rangle].$$

Exercise 2

a) The effective Hamiltonian can be expressed in terms of the Pauli matrices:

$$\tilde{H}_{\text{eff}} = \hbar \begin{bmatrix} -\Delta_L/2 & \Omega/2 \\ \Omega^*/2 & \Delta_L/2 \end{bmatrix} = h_x \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + h_y \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} + h_z \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + h_1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

By equating each element of the matrix we have four equations:

$$\left.\begin{array}{ll} (1): & h_z + h_1 = \frac{-\hbar\Delta_L}{2} \\ (2): & h_x - ih_y = \frac{\hbar\Omega}{2} \\ (3): & h_x + ih_y = \frac{\hbar\Omega^*}{2} \\ (4): & h_1 - h_z = \frac{\hbar\Delta_L}{2} \end{array}\right]$$

By playing with those, we have the following results:

$$(1) + (4) \Longrightarrow h_1 = 0, \quad (2) + (3) \Longrightarrow h_x = \frac{\hbar \text{Re}\,(\Omega)}{2},$$
$$(1) - (4) \Longrightarrow h_z = -\frac{\hbar\Delta_L}{2}, \quad (2) - (3) \Longrightarrow h_y = -\frac{\hbar \text{Im}\,(\Omega)}{2}.$$

Pauli matrices commute each other, so the evolution operator can be written as a separate exponential:

$$e^{-\frac{i\tilde{H}_{\text{eff}}t}{\hbar}} = e^{-\frac{i[h_x\hat{\sigma}_x + h_y\hat{\sigma}_y + h_z\hat{\sigma}_z]t}{\hbar}} = e^{-\frac{i\text{Re}(\Omega)t}{2}} e^{\frac{i\Delta_L t}{2}} e^{\frac{i\text{Im}(\Omega)t}{2}}.$$

The vector precession is:

$$\boldsymbol{h} = \frac{\hbar \text{Re}\,(\Omega)}{2}\boldsymbol{x} - \frac{\hbar \text{Im}\,(\Omega)}{2}\boldsymbol{y} - \frac{\hbar\Delta_L}{2}\boldsymbol{z}.$$

b) Assuming Ω real, the effective Hamiltonian is:

$$\tilde{H}_{\text{eff}} = \frac{\hbar}{2}\boldsymbol{\Omega}_{\text{eff}} \cdot \boldsymbol{\sigma} + \mathbb{I} = \frac{\hbar}{2}\left[\Omega\hat{\sigma}_x - \Delta_L\hat{\sigma}_z\right], \quad \boldsymbol{\Omega}_{\text{eff}} = (\Omega, 0, -\Delta_L).$$

The rotation in the Bloch sphere is governed by the following expression:

$$\boldsymbol{r}(t) = \boldsymbol{r}(0)\cos(\Omega_{\text{eff}}t) + (\hat{\Omega}_{\text{eff}} \times \boldsymbol{r}(0))\sin(\Omega_{\text{eff}}t) + \hat{\Omega}_{\text{eff}}(\hat{\Omega}_{\text{eff}} \cdot \boldsymbol{r}(0))[1 - \cos(\Omega_{\text{eff}}t)].$$

For each case, with the initial atom in the ground state $\boldsymbol{r}(0) = (0, 0, -1)$, we have

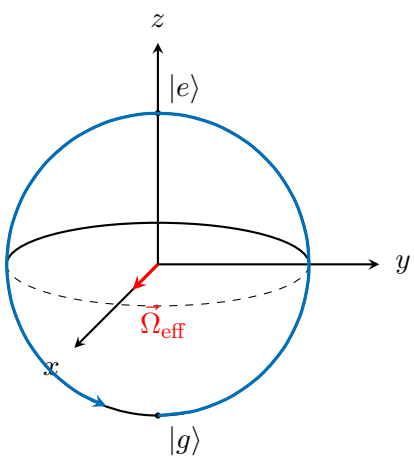
$$\Delta_L = 0: \qquad \boldsymbol{\Omega}_{\text{eff}} = (\Omega, 0, 0), \quad \hat{\Omega}_{\text{eff}} = (1, 0, 0), \quad \Omega_{\text{eff}} = \Omega,$$
$$\Delta_L = \Omega: \qquad \boldsymbol{\Omega}_{\text{eff}} = (\Omega, 0, -\Omega), \quad \hat{\Omega}_{\text{eff}} = \frac{1}{\sqrt{2}}(1, 0, -1), \quad \Omega_{\text{eff}} = \sqrt{2}\Omega.$$

The rotation vector for the first case is:

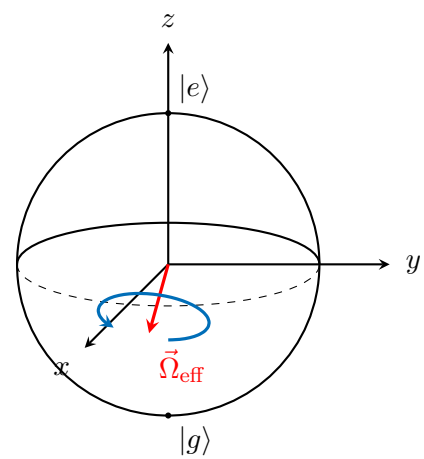
$$\begin{aligned} \boldsymbol{r}(t) &= (0, 0, -1)\cos(\Omega t) + [(1, 0, 0) \times (0, 0, -1)]\sin(\Omega t) + (1, 0, 0)[(1, 0, 0) \cdot (0, 0, -1)][1 - \cos(\Omega t)] \\ &= (0, 0, -1)\cos(\Omega t) + (0, 1, 0)\sin(\Omega t) \\ \boldsymbol{r}(t) &= [0, \sin(\Omega t), -\cos(\Omega t)]. \end{aligned}$$

For the second is:

$$
\begin{aligned}
\boldsymbol{r}(t) =& (0,0,-1)\cos(\sqrt{2}\Omega t) + \frac{1}{\sqrt{2}}[(1,0,-1)\times(0,0,-1)]\sin(\sqrt{2}\Omega t) \\
&+ \frac{1}{2}(1,0,-1)[(1,0,-1)\cdot(0,0,-1)][1-\cos(\sqrt{2}\Omega t)] \\
=& (0,0,-1)\cos(\sqrt{2}\Omega t) + \frac{1}{\sqrt{2}}(0,1,0)\sin(\sqrt{2}\Omega t) + \frac{1}{2}(1,0,-1)[1-\cos(\sqrt{2}t)] \\
\boldsymbol{r}(t) =& \left(\frac{1-\cos(\sqrt{2}\Omega t)}{2}, \frac{\sin(\sqrt{2}\Omega t)}{\sqrt{2}}, -\frac{1+\cos(\sqrt{2}\Omega t)}{2}\right).
\end{aligned}
$$



(a) $\Delta_L = 0$



(b) $\Delta_L = \Omega$